

Efficient Capital Markets: A Review of Theory and Empirical Work

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ABSTRACT

Fama organizes decades of dispersed empirical work on stock-price behavior under a single theoretical roof by formalizing what it means for a market to be efficient. The claim that prices “fully reflect” information is shown to be empirically empty until paired with an assumed model of equilibrium expected returns; the resulting “fair game” property — that no rule conditioned on an information set can earn positive expected excess profit — nests the submartingale model and, as a strictly stronger case, the random walk. Evidence is then organized by information set into weak-form tests (historical prices), semi-strong-form tests (public announcements such as splits and earnings), and strong-form tests (monopolistic access, held by specialists, insiders, and fund managers). Across nearly all of it — serial correlation and filter-rule tests, the Fama-Fisher-Jensen-Roll split study, and Jensen’s risk-adjusted fund-performance measure — Fama finds dependence that is statistically negligible or too small to survive realistic costs, with the exceptions of specialists and, via secondary-offering evidence, insiders. The paper explicitly flags, in its own text, what would later be named the joint-hypothesis problem: every test of efficiency is simultaneously a test of the assumed equilibrium return model.

1 Introduction

Eugene Fama's 1970 *Journal of Finance* paper is not, in form, an empirical study. It is a review article delivered at the 1969 American Finance Association meetings, and its stated task is to organize “the theoretical and empirical literature on the efficient markets model” that had accumulated, largely without a common theoretical vocabulary, since Bachelier's 1900 dissertation. What makes the paper foundational rather than merely a survey is that Fama does not simply catalogue existing results; he first supplies the missing theory that tells the reader which of those results actually bear on market efficiency and which do not.

PAPER Fama's opening move is definitional and, on inspection, destabilizing. He defines an efficient market as one in which security prices “fully reflect” all available information, then states plainly that this definition “is so general that it has no empirically testable implications.” A market can only be tested for efficiency once the process of price formation is specified in enough detail to generate a falsifiable prediction. The paper's entire theoretical section exists to supply that specification, and its entire empirical section exists to report what happens when researchers, mostly working independently of one another and often before Fama's formalization existed, tested the predictions that specification implies.

This review reconstructs that theory in full — the fair-game model, the submartingale special case, and the random walk as a still-stronger special case — because the paper's central methodological claim, and its central vulnerability, both live inside the mathematics. Fama shows that most of the empirical tests economists had been running for seventy years were, technically, tests of the random walk model, which is a strictly stronger and economically unnecessary hypothesis; the weaker fair-game property is what market efficiency actually requires. The review then follows Fama's own three-way partition of the empirical evidence by information set — weak, semi-strong, and strong form — before turning to a critical evaluation that takes seriously a qualification Fama himself states in the paper's own text: that every test of efficiency is, unavoidably, a joint test of efficiency and of whatever equilibrium expected-return model is assumed to hold.

The paper's lasting influence rests less on any single empirical finding than on the taxonomy and the joint-hypothesis caveat themselves — both of which continued to organize, and eventually to complicate, the asset-pricing literature for the following half-century, including work by Fama himself.

2 Research Problem and Hypothesis

2.1 Research Question

Stated precisely rather than descriptively: does the equilibrium expected return on a security, conditioned on progressively larger information sets — first its own price history, then all public information, then all information including that held monopolistically by specialists, insiders, and professional managers — fully absorb that information, such that no trading rule conditioned on

the corresponding information set can generate expected profit in excess of what the assumed equilibrium model already implies? Fama's question is therefore not simply “are markets efficient” but a nested sequence of three narrower, testable questions, one for each information set, unified by a single mathematical criterion (the fair-game property developed in Section 3 below).

2.2 Motivation

The question matters because the “primary role of the capital market,” in Fama's own framing, is the allocation of ownership of the economy's capital stock: firms make production and investment decisions and investors choose among securities under the working assumption that prices are accurate signals. If prices systematically fail to incorporate available information, both sides of that allocation problem are distorted — capital can be misdirected toward projects the market has not correctly evaluated, and investors can be exploited or misled by prices that do not reflect fundamentals. A researcher deciding whether to build models of security valuation on the premise of price accuracy, and a policymaker deciding whether markets need correcting, both need to know how far reality departs from the frictionless ideal.

2.3 Existing Literature as of 1969

Fama frames his own contribution against a specific historical gap. Bachelier (1900) first proposed that speculative price changes should behave as a fair game, a claim Fama notes was essentially ignored for sixty years. Kendall (1953), examining weekly British share-price and commodity indices, found what he described as behavior indistinguishable from cumulative random draws — the “Demon of Chance” passage Fama quotes directly — but supplied no economic rationale for why prices should behave this way. Roberts (1959) drew out the methodological implications of Kendall's finding for financial analysts and, per Fama's own footnote, first proposed the weak-form/strong-form distinction that Fama's taxonomy builds on. Osborne (1959) proposed a Brownian-motion model of price formation with an economic microfoundation Fama explicitly judges too thin (“little in the way of an economic model”). It was only with Mandelbrot (1963) and Samuelson (1965) that the fair-game property received a rigorous economic derivation, via the observation that a futures price equal to the market's conditional expectation of the terminal spot price must follow a martingale.

PAPER Fama's own summary of the state of the field before his paper is unambiguous: “until the Mandelbrot-Samuelson models appeared, there existed a large body of empirical results in search of a rigorous theory.” The random-walk empirical literature and the fair-game theoretical literature had, in effect, developed on separate tracks for most of the preceding decade.

2.4 Research Gap

Two gaps, one theoretical and one organizational, motivate the paper. Theoretically, no existing framework specified precisely which property of prices — independence of successive changes (random walk), non-negative expected return (submartingale), or merely zero conditional expected excess return (fair game) — was actually required by, and actually tested by, the various empirical studies then in circulation; the literature routinely treated random-walk rejections as efficiency rejections, which Fama argues is not generally valid. Organizationally, empirical studies

addressed wildly different questions — serial correlation in daily prices, market reaction to stock splits, mutual-fund performance — without a shared vocabulary connecting them to a single underlying construct.

2.5 Contribution

Fama's claimed contribution is twofold: (i) a general mathematical nesting of the random walk inside the submartingale inside the fair-game/expected-return model, clarifying exactly which empirical test bears on which theoretical claim; and (ii) the weak/semi-strong/strong-form taxonomy (extending Roberts's binary distinction) that organizes the empirical literature by information set. **CRITIQUE** The contribution is real and was field-defining, but it is narrower than it can appear at first reading: Fama does not derive a new equilibrium asset-pricing model, does not generate new data, and — as he states explicitly in the paper's own text (Section 7.1 below) — cannot escape the fact that his organizing framework tests efficiency only jointly with whatever equilibrium return model each cited study happens to assume. The paper's contribution is best understood as a rigorous vocabulary and a synthesis, not as an independent empirical demonstration that markets are efficient.

3 Theoretical Framework and Mechanism

Fama's theory proceeds in three layers of decreasing generality and increasing testability: the expected-return (“fair game”) model, the submartingale model, and the random walk model. Each later layer adds an assumption not required by market efficiency itself, and the paper's central methodological argument is that most of the empirical literature it reviews tests the third, strongest layer, when only the first is economically necessary.

3.1 The Expected-Return, or “Fair Game,” Model

Fama declines to posit a specific equilibrium model (such as the Sharpe-Lintner CAPM) and instead adopts the weakest assumption that gives the efficiency definition testable content: that market equilibrium can be stated in terms of expected returns conditional on an information set.

$$E(\tilde{p}_{j,t+1} | \Phi_t) = [1 + E(\tilde{r}_{j,t+1} | \Phi_t)] \cdot p_{jt} \quad (1)$$

E the expected-value operator.

p_{jt} the price of security j at time t .

$\tilde{p}_{j,t+1}$ its price at $t+1$, inclusive of reinvested intermediate cash income — a random variable as of t .

$\tilde{r}_{j,t+1}$ the one-period percentage return, $(\tilde{p}_{j,t+1} - p_{jt})/p_{jt}$.

Φ_t the information set assumed to be “fully reflected” in the price at t .

Equation (1) operationalizes “fully reflect”: the information in Φ_t is fully utilized in determining the equilibrium expected return, and this — not the vaguer prose definition — is the sense in which Φ_t is impounded in p_{jt} . **PAPER** Fama immediately flags the price of this move: “the assumption that the conditions of market equilibrium can be stated in terms of expected returns elevates the purely mathematical concept of expected value to a status not necessarily implied by the general notion of market efficiency... the results of tests based on this assumption depend to some extent on its validity as well as on the efficiency of the market. But some such assumption is the unavoidable price one must pay to give the theory of efficient markets empirical content.” This single sentence is the paper's own statement of the joint-hypothesis problem, twenty-one years before it acquired that name in Fama's 1991 sequel; it is treated at length in Section 7 below.

Defining the excess market value $\tilde{x}_{j,t+1} = \tilde{p}_{j,t+1} - E(\tilde{p}_{j,t+1} | \Phi_t)$, equation (1) implies directly that

$$E(\tilde{x}_{j,t+1} | \Phi_t) = 0 \quad (2)$$

which, by definition, says that the sequence $\{x_{jt}\}$ is a fair game with respect to $\{\Phi_t\}$: knowing Φ_t is of no help in predicting the sign or magnitude of tomorrow's pricing error. The identical statement holds for the excess (abnormal) return $\tilde{z}_{j,t+1} = \tilde{r}_{j,t+1} - E(\tilde{r}_{j,t+1} | \Phi_t)$, so that $\{z_{jt}\}$ is also a fair game.

The economic content of the fair-game property is a statement about trading systems. Let $\alpha_j(\Phi_t)$ denote the funds a trading system conditioned on Φ_t allocates to security j . The total excess value such a system generates at $t+1$ is $\sum_j \alpha_j(\Phi_t) \cdot \tilde{x}_{j,t+1}$, which by the fair-game property of $\tilde{x}_{j,t+1}$ has expectation exactly zero. **INFERENCE** No mechanical trading rule that conditions only on Φ_t can, in expectation, outperform what the assumed equilibrium model already prices in. This is the precise, falsifiable content that replaces the informal claim that prices “fully reflect” information, and it is the criterion every empirical test in the paper is ultimately measured against.

3.2 The Submartingale Model

A special case of (1) arises when expected returns are assumed non-negative:

$$E(\tilde{p}_{j,t+1} | \Phi_t) \geq p_{jt}, \text{ equivalently } E(\tilde{r}_{j,t+1} | \Phi_t) \geq 0 \quad (3)$$

This is the statement that $\{p_{jt}\}$ follows a submartingale with respect to $\{\Phi_t\}$: next period's expected price, conditional on Φ_t , is at least the current price. Where (3) holds with equality, the sequence is a martingale. The empirical payoff of this special case is sharp and is the theoretical basis for every buy-and-hold benchmark comparison in the paper: for a “one security and cash” trading rule — hold the security, hold cash, or sell short, based only on Φ_t — non-negativity of conditional expected returns directly implies that no such rule can beat a policy of buying and holding the security throughout.

CRITIQUE Fama flags, in a footnote easy to overlook but essential for interpreting every filter-rule test in Section 5, that this buy-and-hold dominance result depends specifically on the non-negativity assumption of the submartingale, not on the more general fair-game model. Under a

general equilibrium model such as Sharpe-Lintner, an individual security's equilibrium expected return can be negative — for instance a security whose return co-moves negatively with the market — in which case cash (with zero return) can legitimately dominate buy-and-hold for that security without any violation of market efficiency. The filter-rule and trading-rule literature reviewed in Section 5 is therefore, strictly, a test of the submartingale special case, not of fair-game efficiency in general; this distinction is rarely made explicit outside the paper itself.

3.3 The Random Walk Model

The strongest and, per Fama, least economically necessary layer requires that the entire conditional distribution of returns, not merely its mean, be independent of Φ_t :

$$f(\tilde{r}_{j,t+1} | \Phi_t) = f(\tilde{r}_{j,t+1}) \quad (4)$$

with the further requirement that f be identical for all t (stationarity). Two separate hypotheses — independence and identical distribution — jointly constitute the random walk model. Contrast this with the weaker restriction the fair-game model actually imposes on the conditional mean alone:

$$E(\tilde{r}_{j,t+1} | \Phi_t) = E(\tilde{r}_{j,t+1}) \quad (5)$$

Equation (4) implies (5) but not the reverse: the random walk asserts the entire distribution is independent of Φ_t , while fair-game efficiency asserts only that the mean is. PAPER Fama's own methodological verdict, central to the paper's argument, is that it is “best to regard the random walk model as an extension of the general expected return or ‘fair game’ efficient markets model,” and that consequently “tests of the ‘random walk’ model that are in fact tests of ‘fair game’ properties are more strongly in support of the model than tests of the additional (and, from the viewpoint of expected return market efficiency, superfluous) pure independence assumption.” The random walk restricts economic behavior more than efficiency requires; higher-moment or higher-order dependence in returns (volatility clustering, for example) is fully compatible with market efficiency provided the conditional *mean* return remains unpredictable.

3.4 Serial Covariance and the Limits of Linear Dependence Tests

To connect the fair-game property to the serial-correlation statistics the empirical literature actually computes, Fama shows that for a fair-game variable $\tilde{x}_t$, $E(\tilde{x}_t \cdot \tilde{x}_{t+\tau}) = 0$ for every lag τ , because $E(\tilde{x}_{t+\tau} | x_t) = 0$ for any subset of Φ_t . CRITIQUE Fama himself immediately supplies a counterexample showing why this result is weaker than it looks: he constructs a three-period binary process in which $x_{t+2} = \text{sign}(x_t \cdot x_{t+1})$, so that all pairwise serial covariances are exactly zero, yet $E(x_{t+2} | x_{t+1}, x_t) \neq 0$ — a trading rule conditioning jointly on the last two observations could earn positive expected profit despite zero linear serial correlation at every lag. Standard serial-correlation and runs tests, which are exactly what the weak-form literature in Section 5 relies on, are therefore structurally incapable of detecting this class of nonlinear dependence. This is an internal limitation the paper flags about its own evidentiary base, not an external criticism.

3.5 The Two-Parameter Model and the Market Model

To extend the fair-game logic from single-security trading rules to a cross-section of securities, Fama reports the Sharpe-Lintner two-parameter equilibrium model as the natural (though, at the time, untested) benchmark:

$$E(\tilde{r}_{j,t+1} | \Phi_t) = r_{f,t+1} + [E(\tilde{r}_{m,t+1} | \Phi_t) - r_{f,t+1}] \cdot \text{cov}(\tilde{r}_{j,t+1}, \tilde{r}_{m,t+1} | \Phi_t) / \sigma^2(\tilde{r}_{m,t+1} | \Phi_t) \quad (6)$$

where $r_{f,t+1}$ is the riskless rate and $r_{m,t+1}$ is the return on the value-weighted market portfolio of all assets. Expected excess return over the riskless rate is proportional to the security's covariance with the market, scaled by the market price of risk $[E(\tilde{r}_{m,t+1} | \Phi_t) - r_{f,t+1}] / \sigma(\tilde{r}_{m,t+1} | \Phi_t)$, a quantity common to every security. PAPER Fama states plainly that “published empirical tests of the Sharpe-Lintner model are not yet available” at the time of writing; the paper substitutes an indirect check, the single-index “market model” originated by Markowitz (1959):

$$\tilde{r}_{j,t+1} = \alpha_j + \beta_j \tilde{r}_{M,t+1} + \tilde{u}_{j,t+1} \quad (7)$$

where r_M is the return on an observable market index (a proxy for the unobservable r_m), and $\beta_j = \text{cov}(\tilde{r}_{j,t+1}, \tilde{r}_{m,t+1} | \Phi_t) / \sigma^2(\tilde{r}_{m,t+1} | \Phi_t)$. FFJR and Blume find the specification well-behaved — stable estimated α_j and β_j over long periods, near-zero serial correlation of residuals and of r_M , and residuals empirically independent of r_M — and Blume's estimated α_j cluster close to zero, broadly consistent with what equation (6) would require after rearrangement, though this is offered by Fama only as indirect, partial corroboration rather than a direct test of Sharpe-Lintner. CRITIQUE This is a genuine gap: nearly every semi-strong and strong-form result in the paper is measured against the statistical market-model residual of equation (7), not against a directly tested equilibrium model, so the entire empirical edifice inherits whatever misspecification the single-index proxy carries.

3.6 The Jensen Performance-Evaluation Benchmark

The paper's clearest applied use of the equilibrium apparatus converts the ex ante relation (6) into an ex post evaluation norm by substituting the realized market return for its expectation:

$$E(\tilde{r}_{j,t+1} | \Phi_t, \tilde{r}_{m,t+1}) = r_{f,t+1} [1 - \beta_j(\Phi_t)] + r_{m,t+1} \beta_j(\Phi_t) \quad (8)$$

Plotting a portfolio's realized return against its estimated β traces a “market line” running from r_f through the market portfolio m ; a portfolio whose (β, return) pair lies above the line has beaten the passive benchmark of matching risk, and below it has underperformed. This vertical deviation is precisely the quantity now known as Jensen's alpha, and it is the sole instrument the paper uses to evaluate strong-form efficiency among professional managers (Section 5.3 below).

3.7 Sufficient (Not Necessary) Conditions for Efficiency

Fama closes the theory section by stating sufficient conditions under which efficiency holds trivially: zero transaction costs, costless information available to all participants, and universal agreement on the implications of information for prices. He is careful to note these are not necessary — transaction costs that inhibit trading do not by themselves prevent prices from reflecting information when trades do occur; disagreement among investors does not imply inefficiency unless some investors can consistently out-evaluate the market. **CRITIQUE** But Fama also concedes that transaction costs, unequal information access, and investor disagreement all “exist to some extent in real world markets,” and states that measuring their effect on price formation “is, of course, the major goal of empirical work in this area.” No formal model is offered of who bears the cost of making prices efficient or why that cost should be borne voluntarily — a gap Grossman and Stiglitz (1980) close a decade later by showing that costless-information efficiency and any incentive to acquire information are jointly impossible in equilibrium (Section 9 below).

4 Data and Empirical Strategy

Because Fama (1970) is a review rather than a primary empirical study, its “data” is the set of studies it synthesizes, and its “empirical strategy” is the weak/semi-strong/strong-form partition itself. Each tier is defined by the information set an investor would need in order to condition a trading rule, and each is tested by a distinct empirical technology developed largely independently across the studies Fama assembles.

Table 1. The three-form taxonomy: information set, test technology, and primary sources reviewed

Form	Information set Φ_t	Test technology	Primary sources
Weak	Historical sequence of prices/returns only	Serial correlation, runs tests, filter rules, spectral analysis, distributional tests	Fama (1965); Alexander; Fama-Blume; Kendall; King; Roll (T-bills)
Semi-strong	All obviously public information (splits, earnings, offerings, rate changes)	Market-model event-study cumulative average residuals	Fama-Fisher-Jensen-Roll (1969); Ball-Brown (1968); Scholes (1969); Waud
Strong	All information, including monopolistically held information	Order-book/transaction analysis; risk-adjusted performance evaluation against the market line	Niederhoffer-Osborne (1966); Scholes (1969); Jensen (1968)

Source: constructed from Fama (1970), Sections III.A–III.C.

The weak-form evidence rests overwhelmingly on Fama's own 1965 dissertation data: daily closing prices for all thirty Dow Jones Industrial Average stocks, running from approximately the end of 1957 to 26 September 1962, yielding roughly 1,200–1,700 daily observations per stock. This sample recurs across the serial-correlation, runs-test, filter-rule (via Fama-Blume), and distributional (stable-versus-normal) analyses reported in Section 5.1 below — a single, fairly narrow cross-

section of large, surviving blue-chip issues carrying most of the weak-form evidentiary weight. The semi-strong evidence is anchored by FFJR's sample of all 940 stock splits on the NYSE from 1927 to 1959 meeting a five-for-four or larger exchange-ratio threshold and at least twelve months of surrounding listing, evaluated through monthly market-model regressions; Ball and Brown's 261-firm, 1946–66 annual-earnings sample; and Scholes's unpublished thesis data on 696 new common-stock issues (1926–66) and an unspecified sample of secondary offerings. The strong-form evidence rests on two much narrower bases: Niederhoffer and Osborne's transaction-level NYSE specialist data, and Jensen's sample of 115 open-end mutual funds over the ten years 1955–64, with the S&P 500 as market-portfolio proxy.

CRITIQUE The empirical strategy's chief methodological vulnerability is visible in this data inventory itself: the weak-form conclusion rests on thirty already-successful, already-listed large stocks over a five-year window, and the strong-form conclusion rests on one specialist study and one mutual-fund study. Neither Fama nor the underlying authors discuss survivorship bias in the DJIA sample, or whether the 940 realized splits and 115 realized funds are representative of the full population of firms and funds that existed, including those that failed, delisted, or were liquidated over the relevant periods. This is not a claim that the results are wrong; it is a claim that the breadth of evidence supporting the paper's confident strong-form conclusion is considerably thinner than the confidence of the prose suggests.

5 Results

5.1 Weak-Form Results

Serial correlation of daily returns on the thirty DJIA stocks is, in Fama's words, "always close to zero" in absolute terms: eleven of thirty first-order coefficients exceed twice their standard error, and twenty-two of thirty are positive, but the average coefficient is only about .03, and Fama's own illustrative calculation shows a coefficient of .06 — itself "more than twice its standard error" — implies a linear relationship that explains only about 0.36 percent of the variance in the next price change. **EVIDENCE** Statistical significance and economic significance diverge sharply here: dependence is real by conventional hypothesis-testing standards but tiny in predictive power. Runs tests corroborate the direction (actual same-sign runs fall short of the expected number for twenty-six of thirty stocks) but the average shortfall is only about three percent of the expected count, and at four- and nine-day differencing intervals the runs tests and serial-correlation tests disagree on the sign of any remaining dependence — evidence, in Fama's assessment, of essentially no dependence beyond the one-day horizon.

Filter-rule tests (Alexander; Fama-Blume, applied to the identical DJIA sample) show that very small filters — 0.5 to 1.5 percent — can generate gross trading profits in excess of buy-and-hold, consistent with the slight positive daily serial correlation just described. But this evidence is decisively reversed once transaction costs are applied: even a New York Stock Exchange floor trader paying only about 0.1 percent per round-trip transaction has this cost more than offset the

small-filter advantage. Alexander's own assessment, quoted directly by Fama, is characteristically blunt: "I should advise any reader who is interested only in practical results, and who is not a floor trader and so must pay commissions, to turn to other sources on how to beat buy and hold."

Distributional evidence (Fama 1965; corroborated by Blume, later extended to Treasury bills by Roll) favors Mandelbrot's non-normal stable (Paretian) class over the Gaussian distribution Bachelier's original model implied: daily return distributions exhibit fatter tails than normal, and — critically for ruling out a simpler finite-variance explanation — do not converge toward normality as the differencing interval lengthens, as a Poisson mixture-of-normals model would predict. Roll's Treasury-bill work supplies the paper's most self-aware illustration of the joint-hypothesis logic: because expected-return variation is not negligible for T-bills (unlike common stocks), testing fair-game efficiency there requires an explicit term-structure model. Roll finds the market-segmentation and liquidity-preference theories fit better than pure expectations, and that once the best-fitting theory is used to compute conditional expected returns, the residual excess-return series is serially independent — efficiency conditional on the assumed term-structure model, precisely the joint test Fama's footnote 24 describes.

5.2 Semi-Strong-Form Results: The FFJR Stock-Split Study

The paper's centerpiece semi-strong evidence is the event study of Fama, Fisher, Jensen and Roll on all 940 NYSE stock splits, 1927–59. A split creates no new claim on real assets, so FFJR's hypothesis is that splits are correlated with, and the market treats them as a signal of, more fundamental information about future dividends and earnings. Using cumulative average market-model residuals across the 940 events, aligned on the split month, and split further by whether the following year brought an above- or below-market dividend change, FFJR find a pattern reproduced schematically in Figure 1.

Schematic pattern of cumulative average residuals around NYSE stock splits

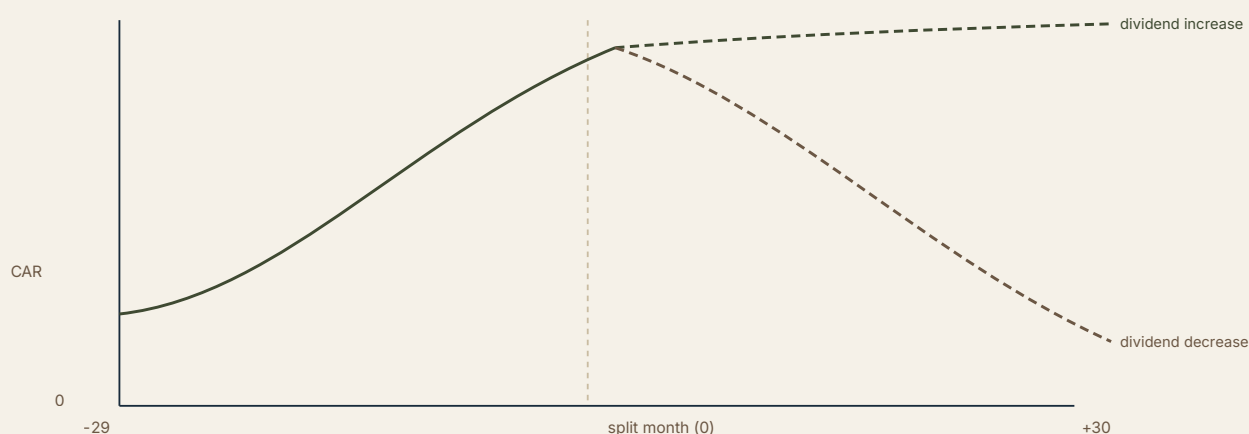


Figure 1. Cumulative average residual (CAR) around the split month, schematic reconstruction

Schematic illustration based on the qualitative pattern reported in FFJR (1969) as described in Fama (1970); the underlying 940-event monthly CAR series was not independently obtainable and this figure does not reproduce the original data points. Not to scale.

Cumulative average residuals rise sharply and almost uniformly across the twenty-nine months before the split — not an artifact of split mechanics, since only about ten percent of splits have more than four months between announcement and effective date, but a reflection of firms tending to split during periods of already-strong, market-relative performance that itself anticipates improved earnings. After the split month, the full-sample residual shows almost no further drift, despite 71.5 percent of split firms (672 of 940) going on to post above-average dividend increases — the market, on average, appears to have priced the anticipated good news in by the split date. The decomposition sharpens this: dividend-increase splits show only a mild further positive drift, while dividend-decrease splits show residuals falling back, once the anticipated increase fails to appear, to roughly the level they held about five months before the split — approximately the earliest point Fama judges reliable split information could plausibly have reached the market. PAPER Fama reports FFJR's own conclusion directly: the results lend "considerable support to the conclusion that the stock market is efficient, at least with respect to its ability to adjust to the information implicit in a split." A footnote qualification is important: the smooth pre-split drift in the *average* residual is not itself a tradeable pattern, because it reflects a few months of abnormally large residuals whose exact timing differs security to security; no individual stock in the sample actually traces the smooth averaged curve.

Other semi-strong evidence corroborates without extending the picture much further. Ball and Brown (1968), studying 261 firms' annual earnings announcements from 1946 to 1966, find cumulative residuals moving in the anticipated direction well before the announcement and conclude that "no more than about ten to fifteen percent of the information in the annual earnings announcement has not been anticipated by the month of the announcement." Waud finds a statistically detectable but economically small (never exceeding half a percent) stock-market reaction to Federal Reserve discount-rate announcements, concentrated in the first trading day, with some evidence of anticipation beforehand. Scholes finds secondary stock offerings associated with a one-to-two-percent decline in cumulative residual return, a decline unrelated to offering size (ruling out a pure selling-pressure explanation) but strongly related to the identity of the seller — largest when the seller is the issuing corporation or a corporate officer — even though that identity is not, at the time of sale, generally public knowledge.

5.3 Strong-Form Results: Specialists and Mutual Fund Managers

Niederhoffer and Osborne's transaction-level NYSE data show reversals (opposite-signed consecutive non-zero price changes) two to three times more frequent than continuations, a pattern the authors trace to the specialist's exclusive access to the limit-order book and the resulting mechanical bunching of orders at round prices. The specialist's resulting profitability is stated in figures Fama quotes directly: the specialist participates as buyer or seller in twenty-four percent of all market volume, sells above his last purchase price on eighty-three percent of his sales, and buys below his last sale price on eighty-one percent of his purchases. Fama's verdict is unambiguous: "the specialist has monopoly power over an important block of information, and, not unexpectedly, uses his monopoly to turn a profit. And this, of course, is evidence of market inefficiency in the strong form sense" — the paper's clearest documented strong-form violation.

Table 2. Jensen (1968): risk-adjusted mutual fund performance relative to the market line, 115 funds, 1955–64

Adjustment	Funds below market line	Average ten-year return deviation
Net returns to investors, including loading charges	89 of 115	–14.6%
Excluding loading (sales) charges	72 of 115	–8.9%
Excluding all published expenses	58 of 115	–2.5%
Also adding back estimated unpublished brokerage commissions	—	+0.09%

Source: Jensen (1968), as reported in Fama (1970), Section III.C.1.

Even in the most favorable adjustment — crediting funds for expenses they did not, in fact, avoid — the average deviation from the market line is 0.09 percent over ten years, which Fama and Jensen both read as economically indistinguishable from zero. Jensen additionally finds no persistence: funds beating the norm in one subperiod are not disproportionately likely to beat it in the next, and the count of funds with large positive deviations is *below* what chance alone would predict under a null of no forecasting ability. Jensen's own conclusion, quoted at length by Fama, treats professional analysts as a maximally favorable test case precisely because they are "extremely well endowed" with resources, making their failure to recover research and transaction costs "a striking piece of evidence in favor of the strong form of the martingale hypothesis... at least as far as the extensive subset of information available to these analysts is concerned."

Fama's own summary verdict for the strong form is correspondingly narrow: efficiency fails demonstrably for specialists, and Scholes's vendor-identity evidence suggests it may fail for corporate insiders and, to a lesser degree, for investment companies acting as sellers in secondary offerings; but "there is no evidence that deviations from the strong form of the efficient markets model permeate down any further through the investment community," making the model "a good first (and second) approximation to reality" for most investors.

6 Robustness and Additional Tests

Concern → Test → Result → Does it resolve the concern? This section evaluates the paper's own robustness checks against that standard.

CONCERN: APPARENT WEAK-FORM DEPENDENCE IS A STATISTICAL ARTIFACT OF THE STANDARD-ERROR FORMULA

Test: Fama notes that because large price changes occur more often than a Gaussian process implies, the conventional $1/\sqrt{N-1}$ standard-error approximation understates the true sampling dispersion of the serial-correlation estimates. Result: this biases significance tests toward over-rejection, meaning some of the "statistically significant" serial correlations reported in Table 1 may be less reliable than a naive reading suggests. Does it resolve the concern? Only partially —

Fama flags the bias but does not re-estimate the standard errors under the stable-distribution assumption his own distributional evidence supports, leaving the exact magnitude of the over-rejection unquantified.

CONCERN: A COMMON MARKET FACTOR COULD MECHANICALLY GENERATE SAME-SIGNED SERIAL CORRELATIONS ACROSS SECURITIES EVEN ABSENT GENUINE DEPENDENCE

Test: cited via King (1966) and Blume, who show a market-wide factor explains roughly half the variance of an individual stock's return. Result: this is consistent with, but does not fully explain away, the pattern of predominantly positive daily serial correlations Fama reports. Does it resolve the concern? No formal correction is applied to Table 1's coefficients for this common-factor contamination; it is noted as a caveat rather than tested directly.

CONCERN: FILTER-RULE PROFITABILITY MIGHT SURVIVE REALISTIC (NON-FLOOR-TRADER) TRANSACTION COSTS

Test: Fama-Blume apply the approximately 0.1 percent round-trip cost faced by an exchange-seat-owning floor trader to their filter-rule gross profits. Result: even this minimal cost erases the small-filter advantage. Does it resolve the concern? Only for the specific investor type tested. **CRITIQUE** Ordinary retail investors of 1970 faced materially higher commissions than 0.1 percent per round trip; the paper does not re-run the filter tests under a realistic retail cost schedule, so the claim that weak-form dependence is universally non-exploitable is demonstrated only for the cheapest possible trader, not for the market participant the efficiency claim is implicitly meant to reassure.

CONCERN: THE FFJR "NO POST-SPLIT DRIFT" FINDING MIGHT SIMPLY REFLECT SMALL-SAMPLE NOISE OR A COINCIDENTAL DIVIDEND-OUTCOME CORRELATION

Test: FFJR decompose the 940-split sample by realized post-split dividend outcome (increase versus decrease) and show the cumulative residual pattern diverges cleanly in the theoretically predicted direction only after the split. Result: the divergence is consistent with market anticipation of the correct dividend outcome, not random noise, since the two subsamples behave identically before the split and diverge only afterward in the direction their subsequent (ex post, then-unknown) dividend news would justify. Does it resolve the concern? Reasonably well for the specific mechanism tested, though the paper does not report standard errors or a formal significance test on the divergence itself, relying instead on the visual pattern of Figure 1.

CONCERN: DISTRIBUTIONAL (FAT-TAIL) EVIDENCE MIGHT INSTEAD REFLECT A FINITE-VARIANCE MIXTURE PROCESS RATHER THAN GENUINE INFINITE-VARIANCE STABLE BEHAVIOR

Test: Fama checks whether return distributions converge toward normality as the differencing interval lengthens, as a Poisson mixture-of-normals model (Press) would predict but a stable-Paretian model would not. Result: no such convergence is found; the high-tailed shape persists (up to scale) across differencing intervals. Does it resolve the concern? Yes, within the alternatives considered — this is one of the paper's more decisive robustness results, directly discriminating between two competing explanations rather than merely restating the original finding.

CONCERN: THE MARKET-MODEL RESIDUAL TECHNIQUE USED THROUGHOUT THE SEMI-STRONG AND WEAK-FORM SECTIONS MIGHT ITSELF BE MISSPECIFIED

Test: FFJR and Blume check the stability of estimated α_j and β_j over long periods and the independence of residuals from the market index and from each other. Result: the specification passes these checks reasonably well, and Blume's estimated α_j cluster near zero as a correctly specified Sharpe-Lintner-consistent model would require. Does it resolve the concern? Not fully — passing internal specification checks (parameter stability, residual independence) is necessary but not sufficient evidence that the market-model proxy correctly captures the true, unobserved Sharpe-Lintner equilibrium; the deeper joint-hypothesis concern (Section 7 below) is untouched by these checks.

7 Critical Evaluation

7.1 The Joint-Hypothesis Problem, in the Paper's Own Words

PAPER This is not a criticism imported from later literature; Fama states it himself, twice, in the paper's own text. In the theory section: "the assumption that the conditions of market equilibrium can be stated in terms of expected returns elevates the purely mathematical concept of expected value to a status not necessarily implied by the general notion of market efficiency... the results of tests based on this assumption depend to some extent on its validity as well as on the efficiency of the market." And again, discussing Roll's Treasury-bill work: "all available tests of market efficiency are implicitly also tests of expected return models of market equilibrium." **CRITIQUE** The paper does not, however, pursue the implication of its own admission. Its confident concluding assessment — that "the evidence in support of the efficient markets model is extensive, and (somewhat uniquely in economics) contradictory evidence is sparse" — is itself necessarily conditional on the largely informal, market-model-based return benchmarks used throughout, several of which (FFJR, Ball-Brown) never test the Sharpe-Lintner model directly at all. A rejection of the joint hypothesis could always, in principle, reflect a wrong equilibrium model rather than a genuinely inefficient market, and the paper's own rhetoric understates how much of its conclusion rests on auxiliary assumptions it explicitly flags but does not test.

7.2 What the Paper Actually Establishes

ESTABLISHED

That daily common-stock returns on large NYSE issues exhibit serial dependence too small, once realistic (floor-trader-level) transaction costs are applied, to generate exploitable trading profit; that the market fully incorporates the earnings-relevant information implicit in a stock split by the split date on average; and that NYSE specialists earn systematic profits from monopolistic access to the limit-order book.

SUPPORTED BUT CONDITIONAL

That professional mutual-fund managers do not, on average, possess exploitable private information — conditional on the Sharpe-Lintner/market-model benchmark used to compute risk-adjusted performance, and on Jensen's estimates of unpublished brokerage costs; and that the Treasury-bill market is fair-game efficient — conditional on Roll's assumed term-structure equilibrium model being correctly specified.

SUGGESTIVE

That corporate insiders and investment companies may possess exploitable private information, based only on the indirect vendor-identity pattern in Scholes's secondary-offering residuals rather than any direct test of insider trading profitability; and that non-normal stable distributions better describe return data than any specific finite-variance alternative, based on the absence of central-limit convergence rather than a positive test discriminating among all plausible fat-tailed alternatives.

NOT ESTABLISHED

That markets are efficient in any sense independent of the equilibrium return model assumed; that the random walk model holds (the paper itself argues this stronger claim is neither well supported nor economically necessary); and that the strong-form conclusion generalizes beyond the two narrow groups (specialists, and by inference from one thesis, insiders) actually studied to the broader universe of potentially informed market participants.

7.3 Strengths

EXTENSION, INDEPENDENTLY IDENTIFIED Beyond the strengths Fama himself claims (the taxonomy, the fair-game/submartingale/random-walk nesting), the paper's disciplined separation of statistical from economic significance is methodologically ahead of much contemporaneous empirical finance — the 0.06-correlation-explains-0.36%-of-variance calculation, and the explicit transaction-cost netting of filter-rule profits, model a standard of interpretation that a great deal of subsequent applied work would not consistently meet. The paper's willingness to report and interpret its own strongest counter-evidence (Niederhoffer-Osborne's specialist findings) as a genuine efficiency violation, rather than explaining it away, also strengthens its credibility as a genuinely critical synthesis rather than an advocacy piece.

7.4 Weaknesses

Problem: No decision-theoretic or information-economics microfoundation for who gathers costly information and why prices should reflect it. **Why it matters:** efficiency is treated as an empirical regularity to be tested rather than an equilibrium outcome to be derived, leaving the paper unable to explain why the frictions it acknowledges exist ("real world markets") should be small enough to leave prices approximately efficient rather than large enough to leave them persistently noisy. **Potential bias:** the paper may overstate how close to the frictionless ideal real markets are,

precisely because it has no model of the equilibrium level of inefficiency such frictions should generate. **How it could be tested:** exactly the exercise Grossman and Stiglitz undertake a decade later (Section 9).

Problem: Survivorship and sample-selection issues are not discussed for any of the three evidentiary bases (thirty already-successful DJIA stocks; 940 realized, non-delisted splits; 115 funds that existed for the full ten-year window). **Why it matters:** funds or firms that failed during the sample period are mechanically excluded, and failure is plausibly correlated with exactly the kind of poor risk-adjusted performance the strong-form tests are designed to detect, which would bias the reported evidence toward finding efficiency. **Potential bias:** the near-zero average mutual-fund underperformance may understate true average underperformance if the worst-performing funds exited before 1964 and are absent from Jensen's sample. **How it could be tested:** reconstruct the full population of funds operating at the start of the period, including those that later closed, and re-estimate average deviation from the market line.

Problem: transaction-cost robustness is calibrated only to the cheapest possible market participant. **Why it matters:** as already noted in Section 6, the paper's efficiency verdict for weak-form dependence rests on a cost assumption far more favorable than what ordinary investors of the period actually faced. **Potential bias:** overstates the practical irrelevance of weak-form dependence for the typical investor. **How it could be tested:** re-run the Fama-Blume filter tests using retail commission schedules contemporaneous with the sample period.

Among these, the single most important weakness is the first: the absence of any equilibrium account of costly information acquisition, because it is the one weakness the paper's own framework cannot even in principle resolve internally — it requires a genuinely different theoretical apparatus, which is exactly what Grossman-Stiglitz would later supply.

8 Contradictions and Edge Cases

When does the submartingale trading-rule result fail even under efficiency? Section 3.2 above showed that buy-and-hold dominance of one-security-and-cash rules depends specifically on the non-negativity of conditional expected returns, not on the weaker fair-game property alone. Under a general equilibrium model, a security with equilibrium expected return below the riskless rate — one whose payoff co-moves negatively with the market — can legitimately be beaten by a rule that substitutes cash during periods it is expected to underperform, with no violation of efficiency whatsoever. Every filter-rule and trading-rule test the paper reviews implicitly assumes this edge case does not bind for the securities tested; for a portfolio of assets with structurally negative risk premia (a case the paper never considers, since all its samples are large-capitalization common stocks with presumptively positive market beta), the entire submartingale-based testing apparatus would misclassify efficient markets as inefficient.

Does zero serial covariance actually establish the fair-game property? No, and the paper proves this about itself. Fama's own three-period binary counterexample (Section 3.4) shows a process with zero pairwise serial covariance at every lag that nonetheless permits positive expected trading profit from a rule conditioning jointly on the last two observations. **INFERENCE** Every weak-

form test in the paper that relies on linear serial correlation or standard runs statistics is, by the paper's own logic, blind to this entire class of nonlinear dependence; the weak-form "pass" the paper reports is therefore a pass against a specific, narrower alternative hypothesis (linear dependence) than market inefficiency in general.

What happens at the extreme of the strong-form benchmark? The strong-form standard — can any group with monopolistic information access earn exploitable profit — produces a stark asymmetry in the paper's own evidence: it fails decisively for specialists (a small, legally privileged group with formal exclusive access to order-flow information) and holds for mutual-fund managers (a large, professionally credentialed but not formally privileged group). **INFERENCE** This pattern is consistent with an interpretation the paper does not itself draw explicitly: strong-form efficiency may hold not because private information cannot be exploited in principle, but because it holds specifically wherever access to superior information is competed for among many similarly resourced participants (mutual fund managers competing against each other) and fails specifically wherever access is legally exclusive to a small group (specialists). This reframes the paper's strong-form conclusion as a statement about the structure of information-access competition, not a general claim about market efficiency under asymmetric information — a reframing that anticipates, without formalizing, the Grossman-Stiglitz mechanism discussed next.

Could another mechanism generate the same semi-strong "no post-split drift" result? An alternative to genuine market efficiency: if firms choose to split specifically when insiders know good news is imminent and about to become public through other channels (the earnings announcement itself, for instance) regardless of the split, the appearance of "prices anticipating the split's information content" could instead reflect prices anticipating the imminent earnings announcement, with the split acting as a correlated but non-causal marker rather than genuine information in itself. FFJR's design, as reported by Fama, does not fully separate these two channels, since both predict the same qualitative CAR pattern.

Does the result survive outside the original setting? The paper's evidence is drawn almost entirely from NYSE-listed, large-capitalization common stocks and Treasury bills during 1926–66. Whether the same conclusions hold for smaller, less liquid securities, for non-U.S. exchanges with different market-maker structures, or for later periods with different institutional arrangements (decimalized pricing, electronic trading, algorithmic market-making) is a question the paper cannot answer from its own evidence — and, as the literature-connections discussion in Section 9 shows, subsequent research (Lo-MacKinlay's rejection of the random walk in size-sorted portfolios; De Bondt-Thaler's long-horizon reversal) finds exactly the kind of dependence the original DJIA-based tests were underpowered to detect once the sample is broadened beyond thirty blue-chip stocks.

9 Further Literature

9.1 Precursors

Louis Bachelier (1900), *Théorie de la Spéculation*. Shared idea: the first formal statement that speculative price changes should constitute a fair game, mathematically equivalent to a martingale. Relationship: Fama credits Bachelier explicitly as the origin of the fair-game intuition underlying the entire theory section, noting the work was ignored for sixty years. Why it matters: establishes that the mathematical core of "efficiency" predates modern finance by seventy years and arose from probability theory rather than economics — useful context for reading Fama's contribution as a synthesis and formalization rather than an original discovery.

Maurice Kendall (1953), "The Analysis of Economic Time-Series, Part I: Prices." Shared idea: the empirical discovery that weekly British share and commodity prices behave like a random walk. Relationship: the direct empirical antecedent Fama quotes at length; Kendall offered no economic rationale, setting up the need for the theory Fama supplies. Why it matters: the empirical regularity preceded any economic explanation of it by over a decade, relevant to any argument that the fair-game theory was constructed to rationalize a pre-existing finding rather than derived independently of it.

Harry Roberts (1959), "Stock-Market 'Patterns' and Financial Analysis." Shared idea: the methodological case for the weak-form/strong-form distinction. Relationship: per Fama's own footnote 1, the direct structural ancestor of the paper's core taxonomy. Why it matters: the weak/strong split often attributed to Fama 1970 is explicitly sourced by Fama himself to Roberts.

M.F.M. Osborne (1959), "Brownian Motion in the Stock Market." Shared idea: a stochastic model of price formation analogous to Bachelier's. Relationship: Fama credits but explicitly criticizes Osborne's economic reasoning as "little in the way of an economic model" — a rare instance of the paper directly critiquing a precursor's rigor. Why it matters: illustrates the pre-1965 gap between statistically sophisticated description and economically rigorous explanation that Mandelbrot and Samuelson, and then Fama, close.

Benoit Mandelbrot (1963), "The Variation of Certain Speculative Prices," and **Paul Samuelson (1965)**, "Proof That Properly Anticipated Prices Fluctuate Randomly." Shared idea: the first rigorous derivations of the fair-game/martingale property from an economic argument (a futures price equal to the market's conditional expectation of the terminal spot price must follow a martingale). Relationship: Fama credits both explicitly, in a dedicated footnote, as the direct theoretical parents of equations (1)–(2) in Section 3 above. Why it matters: Fama's mathematical apparatus is a credited extension of this work, not an independent invention; Mandelbrot's companion contribution, the non-normal stable distribution hypothesis, also directly supplies the paper's distributional evidence (Section 5.1).

9.2 Concurrent and Companion Empirical Papers

Fama, Fisher, Jensen and Roll (1969), “The Adjustment of Stock Prices to New Information.” Shared idea: originates the cumulative-average-residual event-study methodology used as the template for the entire semi-strong section. Relationship: Fama states explicitly that every subsequent semi-strong study reviewed is an "adaptation and extension of the techniques developed in FFJR" — the single most load-bearing methodological citation in the paper. Why it matters: any critique of the semi-strong section (Section 6 above) is implicitly a critique of FFJR's own event-study design.

Michael Jensen (1968), “The Performance of Mutual Funds in the Period 1945–1964.” Shared idea: originates the risk-adjusted alpha performance measure reproduced in Section 3.6. Relationship: the direct primary source for the entire strong-form mutual-fund subsection; Fama quotes Jensen's concluding paragraph verbatim. Why it matters: the paper's strong-form conclusion for professional managers rests almost entirely on this single study.

9.3 Later Critiques and Extensions

Sanford Grossman and Joseph Stiglitz (1980), “On the Impossibility of Informationally Efficient Markets.” Shared idea: formalizes exactly the gap identified in Section 3.7 and Section 7.4 — if information acquisition is costly, prices cannot fully reveal all information in equilibrium, because informed traders would then earn zero return on their costly research and would stop acquiring it; equilibrium requires a positive return to information gathering, hence prices must remain imperfectly, "noisily," efficient. Relationship: directly targets Fama's informal sufficient-conditions discussion by showing that exact efficiency and costly information acquisition are jointly impossible. Why it matters: gives the review's weaknesses section its sharpest theoretical grounding, ten years after the fact, for a gap Fama's own text (Section II.D) implicitly acknowledges but does not formalize.

Michael Jensen (1978), “Some Anomalous Evidence Regarding Market Efficiency.” Shared idea: introduces the term "anomaly" for documented EMH violations and catalogues the earliest examples. Relationship: the same Jensen who supplied Fama 1970's strong-form mutual-fund evidence returns, eight years later, as editor of a dedicated journal issue, to formally catalogue the exceptions accumulating against the model he had helped support. Why it matters: bridges Fama 1970's optimistic "contradictory evidence is sparse" conclusion to the 1980s anomalies literature.

Robert Shiller (1981), “Do Stock Prices Move Too Much to be Justified by Subsequent Changes in Dividends?” Shared idea: a variance-bound test showing stock-price volatility is too high to be consistent with the present-value model under a constant discount rate. Relationship: attacks the same joint-hypothesis structure Fama flags in his own footnote 24, from the opposite direction — excess price volatility rather than return predictability. Why it matters: the exact methodological ambiguity Fama identified in 1970 (efficiency versus the assumed return model) became, a decade later, the central battleground of the volatility-and-predictability debate; later work (including Fama's own) would argue Shiller's result may instead reflect time-varying risk premia rather than inefficiency, illustrating the ambiguity directly.

Andrew Lo and A. Craig MacKinlay (1988), “Stock Market Prices Do Not Follow Random Walks.” Shared idea: re-tests the random walk hypothesis using variance-ratio statistics on weekly CRSP data, 1962–85, and strongly rejects it for aggregate indices and, especially, size-sorted small-stock portfolios. Relationship: an eighteen-years-later methodological upgrade of the weak-form serial-correlation and runs tests reviewed in Section 5.1, with far greater statistical power against exactly the kind of positive autocorrelation the earlier tests were underpowered to detect. Why it matters: complicates Fama 1970's conclusion that weak-form dependence is economically negligible, though whether the dependence Lo-MacKinlay find is exploitable net of costs — Fama's actual efficiency criterion — remains a separate, only partially resolved question.

Werner De Bondt and Richard Thaler (1985), “Does the Stock Market Overreact?” Shared idea: tests whether extreme past losers subsequently outperform extreme past winners over three-to-five-year horizons. Relationship: extends the weak-form testing tradition Fama surveys into a longer horizon than any test the original paper actually reviews, technically still within the weak-form information set (past prices) but empirically unexamined in 1970. Why it matters: the specific anomaly that most directly extends Fama's own weak-form logic into territory his cited literature never explored.

Eugene Fama (1991), “Efficient Capital Markets: II.” Shared idea: Fama's own twenty-one-years-later revision of the taxonomy, replacing weak/semi-strong/strong with tests for return predictability, event studies, and tests for private information, explicitly to accommodate the anomalies literature and to more fully embrace the joint-hypothesis problem as the survey's central organizing theme. Relationship: the direct sequel. Why it matters: lets a reader distinguish what Fama himself later judged needed revision from what external critics identified independently.

Eugene Fama and Kenneth French (1992), “The Cross-Section of Expected Stock Returns.” Shared idea: documents that size and book-to-market, not the CAPM beta of equation (6) above, explain the cross-section of average returns, directly undermining the equilibrium benchmark Fama 1970 treats as the best-available (if untested) norm throughout. Relationship: Fama's own later empirical repudiation of the exact asset-pricing model his 1970 paper used as its implicit standard. Why it matters: the strongest possible illustration, from within Fama's own body of work, of the joint-hypothesis problem he flagged in 1970 — later anomalies could reflect either inefficiency or a misspecified equilibrium model, and Fama-French's own resolution (a multifactor model) is itself just a different auxiliary hypothesis, not a clean test of efficiency alone.

Stephen LeRoy (1989), “Efficient Capital Markets and Martingales.” Shared idea: a rigorous methodological audit arguing that “fair game” terminology conflates several logically distinct properties — martingale, submartingale, random walk — and that much of the empirical literature, including some reviewed by Fama himself, is imprecise about which property is actually being tested. Relationship: the most direct scholarly descendant of Fama 1970's own theory section, revisiting its definitions two decades later in light of the intervening variance-bound and predictability literature. Why it matters: a rigorous secondary check on whether the paper's own mathematical apparatus (Section 3 above) is fully internally consistent.

10 Research Gaps

10.1 Paper-Level Gaps

Gap: no direct test of the Sharpe-Lintner CAPM is available at the time of writing; the multi-security equilibrium benchmark rests entirely on the indirect market-model consistency check of Section 3.5. **Why it matters:** nearly every semi-strong and strong-form conclusion is measured against this untested proxy. **Existing limitation:** Fama states plainly that CAPM tests were still "in progress" in 1970. **Possible way forward:** the direct multi-security CAPM tests that would arrive shortly after (Black-Jensen-Scholes 1972; Fama-MacBeth 1973), which the review's companion Fama-French (1993) entry in this library eventually supersedes with a multifactor alternative.

Gap: the strong-form evidentiary base (specialists, plus one thesis-level insider inference, plus one mutual-fund study) is narrow relative to the sweeping conclusion drawn from it. **Why it matters:** "the efficient markets model seems a good first (and second) approximation to reality" for "most investors" generalizes well beyond the two groups actually studied. **Existing limitation:** no dedicated study of corporate-insider trading profitability is cited (only the indirect Scholes vendor-identity inference). **Possible way forward:** direct insider-trading profitability studies using SEC Form 4 filings, which became available in more systematic form after 1970.

10.2 Literature-Level Gaps (as of 1970, per the paper's own conclusion)

Gap: absence of a well-tested model of equilibrium expected returns under uncertainty. **Why it matters:** Fama identifies this himself as "the most pressing field of future endeavor," directly acknowledging that progress on testing efficiency is bottlenecked by the absence of a validated equilibrium asset-pricing model. **Existing limitation:** the entire semi-strong and strong-form literature substitutes a single-index statistical proxy (the market model) for the untested Sharpe-Lintner equilibrium. **Possible way forward:** the factor-model literature of the following decades, itself never fully closing the joint-hypothesis gap (Section 9.3 above, Fama-French 1992).

10.3 Methodological Gaps

Gap: reliance on single-lag serial correlation and non-parametric runs tests, both of limited power against the kind of positive autocorrelation later shown (Lo-MacKinlay 1988) to be actually present in the data. **Why it matters:** the paper's weak-form "pass" may partly reflect low test power rather than genuine absence of dependence. **Existing limitation:** no variance-ratio or spectral test spanning multiple lags jointly is applied to the DJIA sample. **Possible way forward:** the variance-ratio methodology Lo-MacKinlay introduce eighteen years later.

Gap: no formal model of information-acquisition costs anywhere in the theoretical apparatus. **Why it matters:** without such a model, the paper cannot distinguish "prices reflect information because acquiring and trading on it is costless" from "prices reflect information despite acquisition being costly, for some other equilibrium reason." **Existing limitation:** Section II.D's sufficient conditions are stated but never relaxed formally. **Possible way forward:** Grossman-Stiglitz (1980).

11 Research Opportunities

Idea 1 (Top-Ranked). Quantifying the joint-hypothesis wedge using modern factor models

RESEARCH QUESTION — How much of the weak-form dependence documented by Lo-MacKinlay (1988) and later authors survives once returns are risk-adjusted using a modern multifactor model instead of Fama 1970's implicit constant-expected-return benchmark?

MOTIVATION — Operationalizes Fama's own footnote 24 admission using tools unavailable in 1970.

HYPOTHESIS — A meaningful share of documented "weak-form dependence" reflects time-varying risk premia rather than genuine unpredictability failures.

ECONOMIC MECHANISM — Expected returns vary with priced risk factors (value, momentum, liquidity); naive serial-correlation tests assume a constant expected return and misattribute risk-premium variation to "dependence."

LITERATURE CONNECTION — Fama 1970 (joint hypothesis), Fama-French 1992/1993, Lo-MacKinlay 1988.

DATA — CRSP daily/monthly returns, 1926–present; Fama-French factor data (Ken French data library, freely available).

METHOD — Re-run serial-correlation, runs, and filter tests on multifactor-model residuals rather than raw returns; compare Sharpe ratios of any residual predictability, net of realistic costs.

IDENTIFICATION STRATEGY — Difference-in-tests design comparing raw-return versus residual predictability across sub-periods and asset classes.

EXPECTED CONTRIBUTION — A direct, quantitative decomposition of "efficiency violation" into risk-based and genuinely anomalous components.

MAIN RISK — Multifactor models are themselves an auxiliary equilibrium hypothesis, so the joint-hypothesis problem is narrowed, not eliminated.

Idea 2 (Top-Ranked). High-frequency replication of the Niederhoffer-Osborne reversal pattern

RESEARCH QUESTION — Do the transaction-level reversal patterns Niederhoffer-Osborne documented on the 1960s NYSE persist in modern limit-order-book equity and crypto markets, and are they still non-exploitable net of costs?

MOTIVATION — Fama treats transaction-level dependence as mechanical and irrelevant to efficiency; algorithmic trading and colocation may have changed that calculus entirely.

HYPOTHESIS — Reversal patterns persist qualitatively, but the exploitability threshold has shifted because marginal transaction costs have fallen by orders of magnitude.

ECONOMIC MECHANISM — Limit-order-book bunching versus modern maker-taker fee structures and high-frequency market-making.

LITERATURE CONNECTION — Niederhoffer-Osborne 1966 (via Fama 1970 Section III.A.3), the submartingale trading-rule framework of equation (3).

DATA — Millisecond-level NYSE TAQ/LOBSTER order-book data; comparable crypto-exchange order-book data via public APIs.

METHOD — Replicate the original reversal-frequency statistics at tick-by-tick frequency; estimate realistic round-trip net profitability.

IDENTIFICATION STRATEGY — Within-market comparison across microstructure regime changes (Reg NMS, decimalization) and across market types.

EXPECTED CONTRIBUTION — A direct test of whether Fama's submartingale benchmark holds at the frequency where dependence is strongest, at data resolution unavailable in 1970.

MAIN RISK — Results are highly sensitive to assumed latency and fee structure, which vary substantially by trader type.

Idea 3 (Top-Ranked). Machine-learning anomaly detection as a modern strong-form test

RESEARCH QUESTION — Can a model trained only on public information generate risk-adjusted excess returns, and how quickly do such signals decay — a continuous-time analogue to Fama's discrete event studies?

MOTIVATION — Fama's semi-strong tests examine only a handful of discrete information types; modern alternative data permits testing a far larger information set for incorporation speed.

HYPOTHESIS — Public-information-only machine-learning strategies generate statistically detectable but economically small, rapidly decaying abnormal returns.

ECONOMIC MECHANISM — Information-processing costs create a temporary edge for sophisticated processors, but competition among similarly equipped quant firms compresses it toward zero net of costs, echoing Fama's "sufficient numbers of investors" condition.

LITERATURE CONNECTION — Fama 1970 semi-strong section, Grossman-Stiglitz 1980, the Jensen-alpha framework of Section 3.6.

DATA — SEC EDGAR filings, earnings-call transcripts, news-sentiment data; CRSP/Compustat for returns and fundamentals.

METHOD — Train and out-of-sample test models on rolling windows; compute Jensen-alpha-style abnormal returns and signal decay half-life.

IDENTIFICATION STRATEGY — Strict walk-forward, no-look-ahead validation; benchmark against realistic implementation costs.

EXPECTED CONTRIBUTION — Quantifies the modern speed of adjustment to new information at a resolution Fama's monthly event studies could not measure.

MAIN RISK — Overfitting and data-snooping with large alternative datasets risk reproducing exactly the data-mining concern the original paper never addresses.

Idea 4. The three-form taxonomy applied to cryptocurrency markets

RESEARCH QUESTION — Do weak-, semi-strong-, and strong-form efficiency hold in major cryptocurrency markets, and does the taxonomy transfer to a market with no dividend/earnings equivalent and highly concentrated informed participants?

MOTIVATION — Crypto market structure differs from 1960s NYSE equities in ways that plausibly change every mechanism Fama discusses.

HYPOTHESIS — Weak-form dependence is larger and more persistent than in Fama's equity evidence; semi-strong adjustment to listing and protocol-upgrade announcements is slower.

ECONOMIC MECHANISM — Lower institutional arbitrage capital and less mature market-making should, per the submartingale logic of equation (3), permit larger economically exploitable profits before being competed away.

LITERATURE CONNECTION — Fama 1970 Sections III.A and III.B; Grossman-Stiglitz 1980.

DATA — Public exchange APIs for tick/daily price data since approximately 2013; on-chain data for listing and upgrade event dates.

METHOD — Replicate Fama's serial-correlation, runs, and filter-rule tests; replicate the FFJR event-study design around listings and upgrades.

IDENTIFICATION STRATEGY — Cross-sectional comparison across tokens of varying market capitalization and age.

EXPECTED CONTRIBUTION — Tests whether Fama's conclusions are a property of 1960s U.S. equity institutions or a more general feature of organized markets.

MAIN RISK — Extreme survivorship bias and wash-trading concerns could contaminate both weak-form and event-study estimates.

Idea 5. Modern robust-inference re-audit of the FFJR event-study methodology

RESEARCH QUESTION — Does the FFJR/Fama 1970 cumulative-average-residual methodology, re-run with modern robust standard errors and a multifactor benchmark, still support the "information fully incorporated by announcement" conclusion for stock splits?

MOTIVATION — FFJR's original inference relies on classical OLS residuals from a single-factor market model, with no correction for cross-sectional correlation of event-time residuals or multiple-testing concerns.

HYPOTHESIS — The core "no post-split drift" finding survives, but the significance of some sub-period effects (the dividend-decrease reversal) is overstated under classical inference.

ECONOMIC MECHANISM — Unchanged from the original study; the question is inferential, not mechanistic.

LITERATURE CONNECTION — FFJR 1969, Fama 1970 Section III.B, modern event-study methodology.

DATA — CRSP daily/monthly split data extended to NASDAQ and the present; Ken French factor data.

METHOD — Replicate the cumulative-average-residual construction under both the original and a multifactor specification; compute Fama-MacBeth, clustered, and bootstrap standard errors.

IDENTIFICATION STRATEGY — Like-for-like replication varying only estimation window, benchmark model, and inference method.

EXPECTED CONTRIBUTION — Connects Fama 1970's foundational event-study evidence to the modern econometric literature on event-study inference.

MAIN RISK — Pre-1970 split data quality may not meet modern cleaning standards, complicating exact replication.

11.1 Ranking

The three ideas ranked above the ordering line (1–3) are judged strongest on the combination of novelty, feasibility, and economic importance. Idea 1 ranks highest because it directly operationalizes the paper's own stated central limitation using data and factor models that did not exist in 1970, at essentially zero data-acquisition cost (CRSP and the Ken French library are both standard, freely available academic resources) and with a clear, falsifiable output. Idea 3 ranks second: it extends the semi-strong-form research program to a genuinely new information technology and is highly feasible given modern NLP and alternative-data infrastructure, though its main risk (data snooping) is real and must be designed against carefully. Idea 2 ranks third: it is the most direct like-for-like replication of a specific finding in the paper (Niederhoffer-Osborne) using data of a kind Fama could not have imagined, but it is more data- and infrastructure-intensive than Idea 1, and its economic conclusions are more sensitive to assumptions about trader-specific cost structures. Ideas 4 and 5, while both worthwhile, are ranked lower: Idea 4 faces serious data-quality confounds specific to crypto markets that are harder to control for than the confounds facing Ideas 1–3, and Idea 5 is valuable but primarily methodological rather than substantively novel.

12 Learning Framework

12.1 Prerequisite Concepts

Conditional expectation and the tower property; the definition of a martingale and submartingale with respect to a filtration (information sequence); ordinary least squares regression and the interpretation of residuals; the Sharpe-Lintner capital asset pricing model in its basic form; the distinction between the normal and stable (Paretian-Lévy) distribution families; and the elementary logic of an event study (cumulative average residuals around a corporate action).

12.2 Core Concepts Learned

That market efficiency, formalized, is a statement about conditional expected returns rather than about price predictability in general; that the random walk model is a strictly stronger and economically unnecessary special case of efficiency; that every empirical test of efficiency is unavoidably a joint test of efficiency and of an assumed equilibrium return model; and that weak, semi-strong, and strong efficiency correspond to progressively larger information sets, each requiring a distinct empirical technology to test.

12.3 Important Equations

The fair-game/expected-return model, $E(\tilde{p}_{j,t+1}|\Phi_t) = [1 + E(\tilde{r}_{j,t+1}|\Phi_t)] \cdot p_{jt}$ (equation 1); the submartingale condition $E(\tilde{r}_{j,t+1}|\Phi_t) \geq 0$ (equation 3) and its buy-and-hold trading-rule implication; the random walk's density equality $f(\tilde{r}_{j,t+1}|\Phi_t) = f(\tilde{r}_{j,t+1})$ (equation 4); the Sharpe-Lintner expected-return relation (equation 6) and its single-index market-model proxy (equation 7); and the Jensen performance-evaluation benchmark (equation 8), the direct ancestor of what is now universally called Jensen's alpha.

12.4 Methods Learned

The cumulative-average-residual event-study technique; filter-rule and runs-test methodology for weak-form dependence; the use of a single-index market-model regression as a practical proxy for an untested multi-factor equilibrium model; risk-adjusted performance evaluation against a market-line benchmark; and the general methodological discipline of separating statistical significance from economic (transaction-cost-adjusted) significance.

12.5 Questions to Test Understanding

COMPREHENSION

Can you state, in your own words, why the claim that prices "fully reflect" information has no testable content on its own, and what additional assumption Fama introduces to make it testable? Can you explain the difference between the fair-game, submartingale, and random walk models, and identify which one each empirical test in the paper actually tests?

ANALYSIS

Why does buy-and-hold dominance of one-security-and-cash trading rules require the submartingale's non-negativity assumption specifically, rather than following from the general fair-game model? Why is Fama's own three-period counterexample (Section 3.4) a genuine limitation of the entire weak-form evidentiary base, rather than a minor technical aside?

RESEARCH

If you were to redesign Jensen's mutual-fund performance test using a modern multifactor benchmark instead of the single-index market model, what result would you expect, and why might it differ from Jensen's original near-zero finding? Can you construct a market setting, consistent with Fama's own sufficient conditions in Section 3.7, in which the strong form fails for one class of informed investor but not another — and does the specialist-versus-mutual-fund-manager asymmetry documented in the paper already constitute such a case?

12.6 Papers to Read Next

Fama, Fisher, Jensen and Roll (1969), the primary source for the entire event-study methodology this paper only summarizes; Jensen (1968), the primary source for the strong-form mutual-fund evidence and the Jensen-alpha measure; Grossman and Stiglitz (1980), the essential theoretical counterweight formalizing the information-cost gap left informal here; Fama (1991), Fama's own twenty-one-years-later revision of this exact framework; and Lo and MacKinlay (1988), the most direct methodological upgrade of the weak-form tests reviewed in Section 5.1.

13 Conclusion

One-sentence thesis: the claim that security prices "fully reflect" available information becomes empirically testable only once paired with an assumed model of equilibrium expected returns, and once so paired, the evidence assembled from 1900 through 1970 supports efficiency with respect to historical prices and public announcements, but not with respect to the private information held by exchange specialists.

Three most important insights. First, the random walk model long treated as synonymous with market efficiency is in fact a strictly stronger and economically unnecessary hypothesis; the weaker fair-game property, requiring only that conditional expected returns be unpredictable, is what efficiency actually demands, and it is far better supported by the evidence than the random walk itself. Second, every test of market efficiency is unavoidably a joint test of efficiency and of whatever equilibrium return model is assumed — a limitation Fama states explicitly in his own text, twenty-one years before his own 1991 sequel gave it a name. Third, the paper's strong-form evidence reveals a sharp and under-examined asymmetry: efficiency fails decisively wherever information access is legally exclusive to a small group (specialists) and holds wherever superior information must be competed for among many similarly resourced participants (mutual fund managers).

Strongest evidence. The FFJR stock-split event study: across 940 NYSE splits from 1927 to 1959, cumulative average residuals rise before the split in anticipation of favorable dividend news and show almost no further drift afterward, on average, despite most split firms subsequently raising dividends — a large, carefully constructed sample showing information incorporation essentially complete by the public announcement date.

Biggest weakness. The absence of any equilibrium account of costly information acquisition: the paper treats efficiency as an empirical regularity to be measured rather than an outcome to be derived from an explicit model of who gathers information, at what cost, and why competition should drive that cost's return toward zero without eliminating the incentive to gather information in the first place — the gap Grossman and Stiglitz would formalize a decade later.

Most important research gap. The absence, at the time of writing, of any validated multi-security equilibrium model of expected returns under uncertainty, which Fama himself identifies as "the most pressing field of future endeavor" and which every semi-strong and strong-form conclusion in the paper implicitly depends on.

Best research opportunity. Using modern multifactor asset-pricing models to directly decompose documented weak-form dependence (Lo-MacKinlay 1988 and successors) into a risk-based component and a genuinely anomalous residual, operationalizing the exact joint-hypothesis wedge Fama's own footnote 24 identifies but does not measure.

Three papers to read next. Fama, Fisher, Jensen and Roll (1969); Grossman and Stiglitz (1980); and Fama (1991).

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